Point-slope form

1 Slope-intercept form

2 $y - y_1 = m(x - x_1)$

3 Yes

4

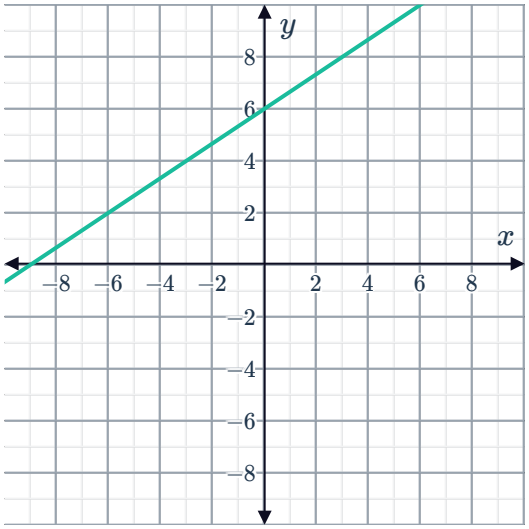
a

i $b = 6$

ii $y = \frac{2x}{3} + 6$

iii $x = -9$

iv



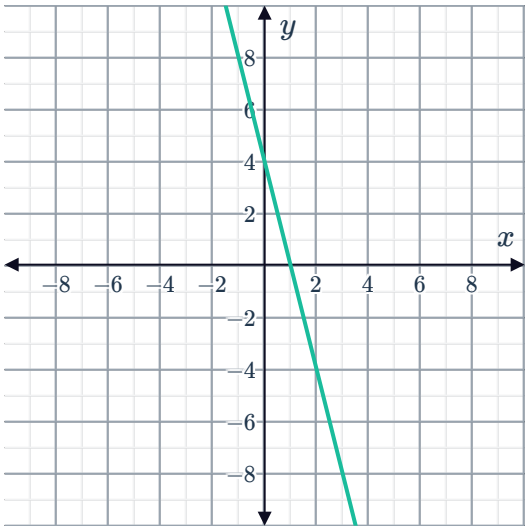
b

i $b = 4$

ii $y = -4x + 4$

iii $x = 1$

iv



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5 **a** $b = 7$

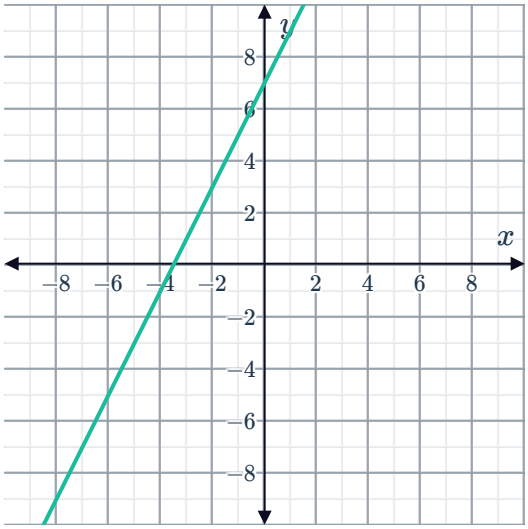
b $y = 2x + 7$

c $y - 5 = 2\,(x - (-1))$

$y - 5 = 2x + 2$

$y = 2x + 7$

d



6

a $10x + y = -1$

b No

7

a $9x + 2y = 17$

b $5x - 9y = -50$

8

a No **b** Yes **c** No **d** Yes

9

a

i 4 **ii** $y = 4x + 5$

b

i 7 **ii** $y = 7x - 9$

c

i -4 **ii** $y = -4x + 3$

d

i 9 **ii** $y = 9x + 8$

e

i 5 **ii** $y = 5x - 6$

f

i $\frac{4}{3}$ **ii** $y = \frac{4}{3}x - 5$

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g

i 4

ii $y = 4x - 1$

h

i $-\frac{13}{7}$

ii $y = -\frac{13x}{7} - \frac{23}{7}$

10

a $\frac{13}{6}$

b $13x - 6y = 15$

11

a $y = -2x - 13$

b $y = 2x - \frac{27}{4}$

c $y = -\frac{51}{2} + \frac{9x}{2}$

d $y = 3x + 10$

e $y = 4x + 25$

f $y = 3x - \frac{15}{4}$

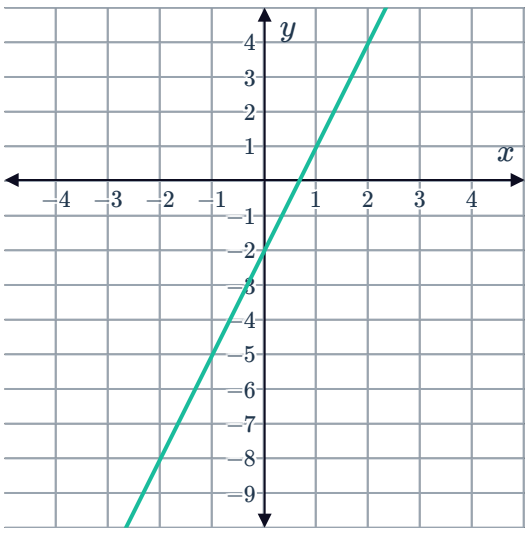
g $y = \frac{3x}{2} + \frac{33}{2}$

Additional questions

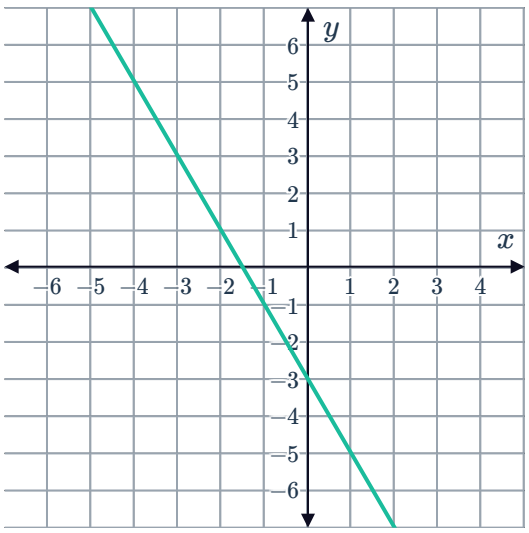
12 $y + 3 = -\frac{4}{3}(x - 2)$

13

a

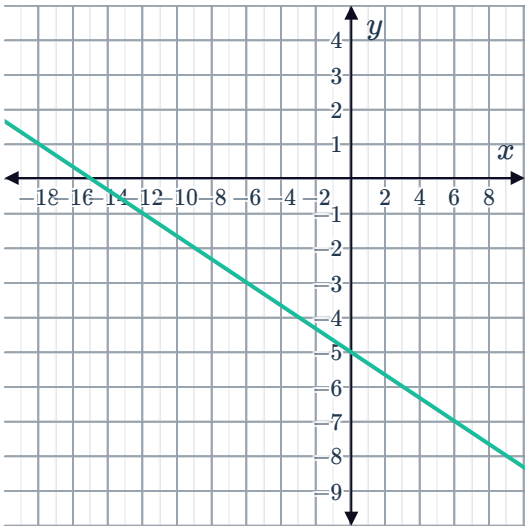


b

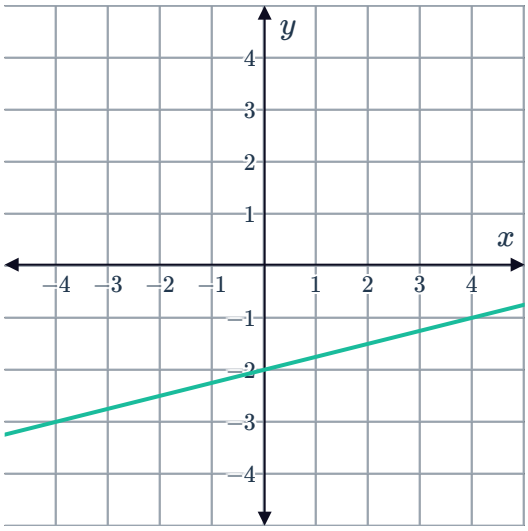


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c



d



14

a $(y - 665) = 8(x - 80)$

b $y = 8x + 25$

c $\$25$

d Domain: $x \in [50, 100]$

15

a 500 mph

b $y - 1042 = 500\,(x - 2)$

c $x = 11$

16

a A scenario where we are given a rate of change (slope) and both variables at a particular moment (point). For example, if we are told how much someone deposits into their savings account per week and the amount in their bank account after a certain number of weeks.

b A scenario where we are given the rate of change (slope) and the initial value (y-intercept). For example, if we are told how much a plumber charges as their flat callout fee (y-intercept) and how much they charge per hour (slope).

c A scenario where we are given the unit rates for two items that we are creating a mixture of to get a certain total. For example, if we are told the cost of almonds per pound (A), the cost of peanuts per pound (B), and are told the total amount to be spent (C).

17 Point-slope form: $y - y_1 = m\,(x - x_1)$
Slope-intercept form: $y = mx + (y_1 - mx_1)$
The variable m in both forms represents the slope of the line. The y-intercept of a line with slope m and point (x_1, y_1) is $(0, y_1 - mx_1)$.

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18

a $y - 15 = 2(x - 4)$

b $y = 2x + 7$

c y-intercept$= 7$
This means that there was $= 7$ oz of water in the cup before it was placed under the leaking faucet.

d The cup will overflow after 5.5 hours.
Domain: $0 \leq x \leq 5.5$
Range: $7 \leq y \leq 18$

19

a Domain: $x \in [10, 100]$
This means that Paige must have a minimum of 10 people and a maximum of 100 people at the event.

b This is not correct. She calculated the rate of change from $(100, 1625)$ to $(0, 0)$, but there is an initial fee as we can see if we were to extrapolate to 0 people. If we instead calculate the cost from $(10, 500)$ to $(100, 1625)$, we get a unit cost of $\$12.50$ per person.

c $y - 500 = 12.5(x - 10)$

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